ALI AHMED GAD AHMED

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PROFESSIONAL SUMMARY

Systems & Biomedical Engineering student at Cairo University with hands-on experience in Artificial Intelligence, Backend Development, and Data Engineering. Skilled in developing scalable applications, data-driven solutions, and machine learning systems. Strong analytical, problem-solving, and teamwork abilities demonstrated through multiple technical projects. Local Winner and Global Nominee at the NASA Space Apps Challenge 2025. Eager to contribute to impactful engineering projects and gain industry experience.

EDUCATION

Cairo University — Giza, Egypt
B.Sc. in Systems & Biomedical Engineering, Cairo University
Expected Graduation: 2027 | GPA: 3.56/4.0

Modern Narmer Language School — Giza, Egypt
High School Diploma (A+) | 2023

EXPERIENCE

Data Engineering Trainee — DEPI Microsoft Data Engineering Track | Nov 2025 – Present

• Designed and implemented end-to-end data engineering solutions using Python, PySpark, Apache Kafka, Databricks, Neo4j, Snowflake, and PostgreSQL.
• Built a capstone project featuring scalable ETL pipelines, real-time data streaming, cloud data warehousing, and analytics dashboards.
• Delivered **3 freelance** data engineering projects, developing production-ready data pipelines and automated workflows for real-world business requirements.
• Applied distributed data processing techniques to handle large-scale datasets, improving data accessibility, processing efficiency, and reporting capabilities.
• Collaborated within agile teams to design, optimize, and deploy cloud-native data solutions following industry best practices.

Machine Learning Engineer — NASA Space Apps Challenge 2025 | Oct 2025

Local Winner & Global Nominee

• Developed an AI-powered geospatial analytics platform utilizing NASA satellite datasets to identify and predict environmental hotspots.
• Designed and trained Random Forest machine learning models, achieving accurate environmental risk classification and prediction.
• Led the development of real-time environmental monitoring and visualization features to support data-driven decision-making.
• Selected as a Local Winner among **300+** competing teams, advancing to the Global Nominee stage of the world's largest international space and science hackathon.
• Awarded **EGP 10,000** and professional technical training scholarships in recognition of project innovation and impact.

PROJECTS

DataDose – Drug Interaction Intelligence Platform | Jan 2026 – June 2026

• Developed a scalable healthcare intelligence platform for drug interaction analysis using PySpark, Apache Kafka, Databricks, Neo4j, Snowflake, and FastAPI.
• Processed, cleaned, and standardized **25,000+** pharmaceutical records from FDA and pharmacy datasets, unifying drug names, active ingredients, and therapeutic classifications.
• Designed a real-time data engineering pipeline integrating Kafka, Databricks, Snowflake, and Neo4j to

support large-scale pharmaceutical data processing and analytics.
• Built a graph database modeling complex drug-to-drug relationships and interaction networks, enabling efficient querying and knowledge discovery.
• Deployed analytics services through a web-based platform for pharmacists and healthcare professionals using industry-standard, cloud-native technologies.

EU App – Fitness & Rehabilitation Healthcare Platform | Feb 2026 – May 2026

• Developed a healthcare platform supporting personalized nutrition planning, fitness tracking, rehabilitation programs, and chronic disease management.
• Collected and integrated healthcare data by web scraping **3** external platforms, ingesting **23,000+** food recipes, **2,800+** fitness workouts, and **531+** rehabilitation exercises into a cloud-hosted database.
• Designed scalable PostgreSQL/Supabase database architectures and implemented secure backend services using FastAPI and SQLAlchemy.
• Developed and documented **20+** RESTful API endpoints supporting authentication, exercise management, nutrition planning, and healthcare analytics workflows.
• Applied data cleaning, transformation, and validation pipelines to improve data quality and support production-scale application requirements.

Arabic Dialect Identification System | May 2026

• Developed an AI-powered speech classification system capable of identifying **4** major Arabic dialects using machine learning and acoustic feature engineering techniques.
• Trained and optimized an XGBoost model on speech datasets from Hugging Face, utilizing MFCC-based feature extraction, preprocessing, and hyperparameter tuning.
• Achieved **90%** classification accuracy, with strong precision and recall performance across multiple dialect categories.
• Integrated speech transcription, translation, and dialect-conversion services into a full-stack application using FastAPI and React.
• Designed an end-to-end machine learning pipeline covering data preprocessing, model training, evaluation, and real-time inference.

TECHNICAL SKILLS

Programming & Databases: Python, SQL, Java, PostgreSQL, C++
Data Engineering: PySpark, Apache Kafka, Databricks, Snowflake, Neo4j, ETL Pipelines
AI & Machine Learning: PyTorch, TensorFlow, Scikit-learn, XGBoost, Deep Learning, CNNs, RNNs
Backend Development: FastAPI, SQLAlchemy, REST APIs, Supabase, Microsoft Azure
Data Analysis & Processing: Pandas, NumPy
Tools & Methodologies: Git, GitHub, Jira, Agile/Scrum

CERTIFICATIONS & TRAINING

• Microsoft Data Engineering & Big Data Professional Training Program (DEPI)
• Machine Learning Specialization – DeepLearning.AI & Coursera (Instructor: Andrew Ng)

LANGUAGES

Arabic – Native | English – Fluent (B2)